



WORKSHOP 2026

AI-Assisted Development

Build Modern Applications Like a Startup

Team Stero Sonic Labs



ONE QUESTION

How are modern applications actually built today?

Today you'll learn the complete workflow.

Why This Workshop?

Traditional education teaches

- ✗ Programming
- ✗ DSA
- ✗ DBMS

Industry expects

- ✓ AI-assisted development
- ✓ Product thinking
- ✓ Deployment & APIs
- ✓ Cloud
- ✓ AI

This workshop bridges that gap.

About Stero Sonic Labs



AI Applications

Intelligent solutions
for real problems



Automation

Workflows that
run themselves



Modern Web Apps

Fast, scalable,
beautiful



Product Engineering

From idea to
launch



AI Agents

Autonomous
digital workers

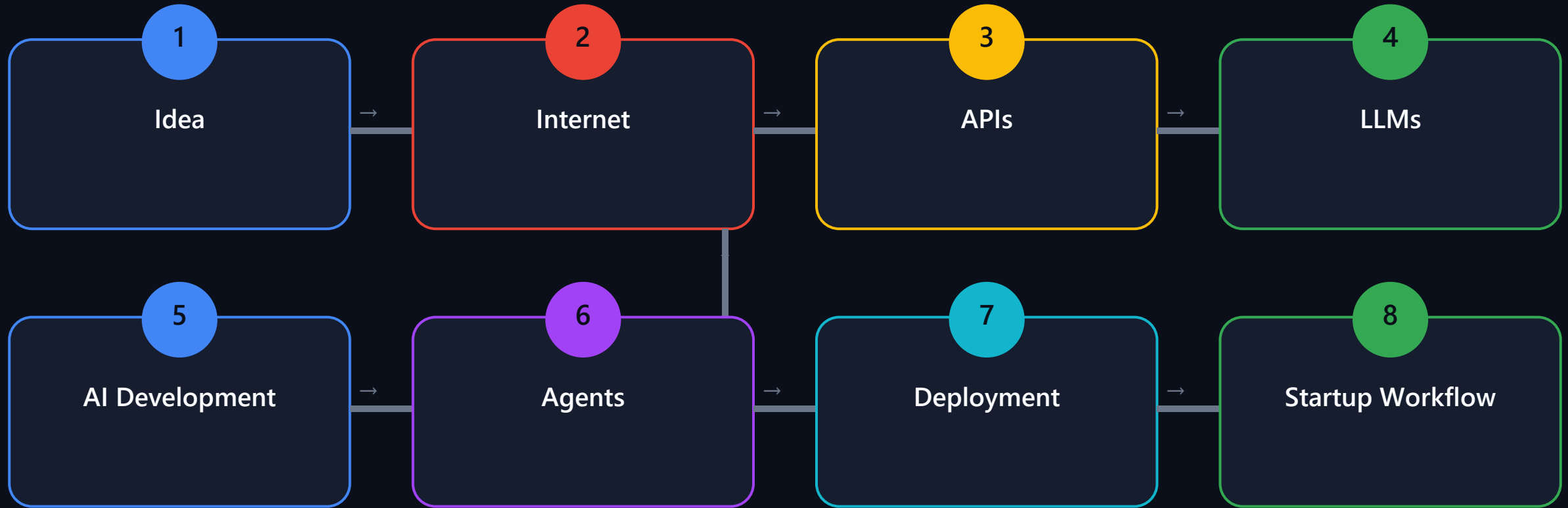


Cloud Solutions

Infrastructure
that scales

Roadmap

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SECTION 2

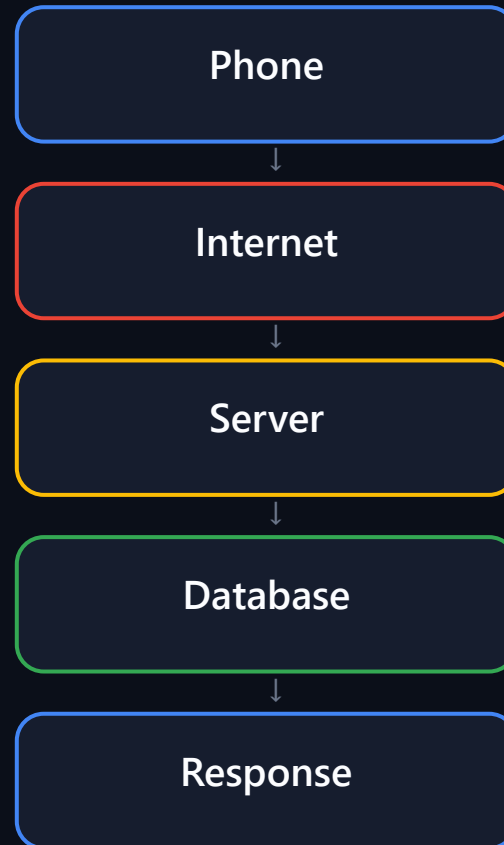
How Applications Work

The journey of a request — from your phone to the database and back.



How Does Instagram Load Your Feed?

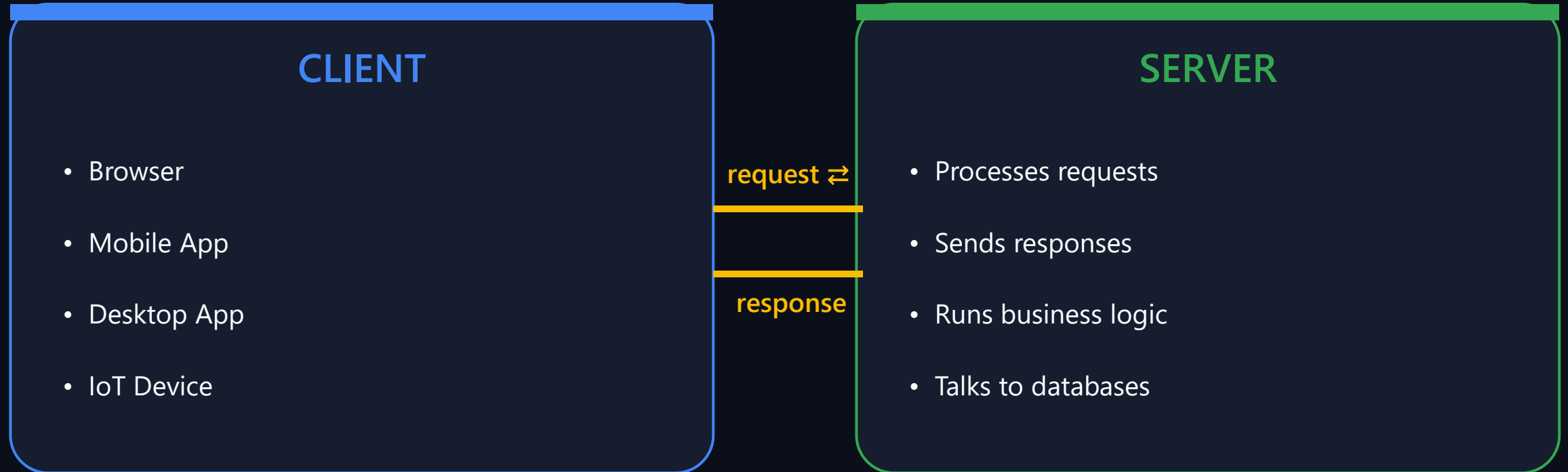
How does your phone receive data?



⚡ What's happening?

1. Your phone asks Instagram
2. Data travels over the internet
3. Servers process the request
4. Database returns your feed
5. Response comes back — rendered!

Client vs Server



Frontend

User Interface

Everything the user sees and touches

What runs in the browser

- HTML — structure
- CSS — styling
- JavaScript — behavior
- React — component framework



Your code becomes pixels

Backend

Business Logic

The rules of your product — payments, orders, matching

Authentication

Who are you? Verify identity & sessions

APIs

Expose functionality over HTTP for the frontend

Database Access

Read/write data safely and efficiently

Database

Stores application data

- ✓ Persistent — survives restarts
- ✓ Queryable — find data fast
- ✓ Consistent — ACID guarantees
- ✓ Scalable — grows with users

SQL vs NoSQL

SQL — structured tables

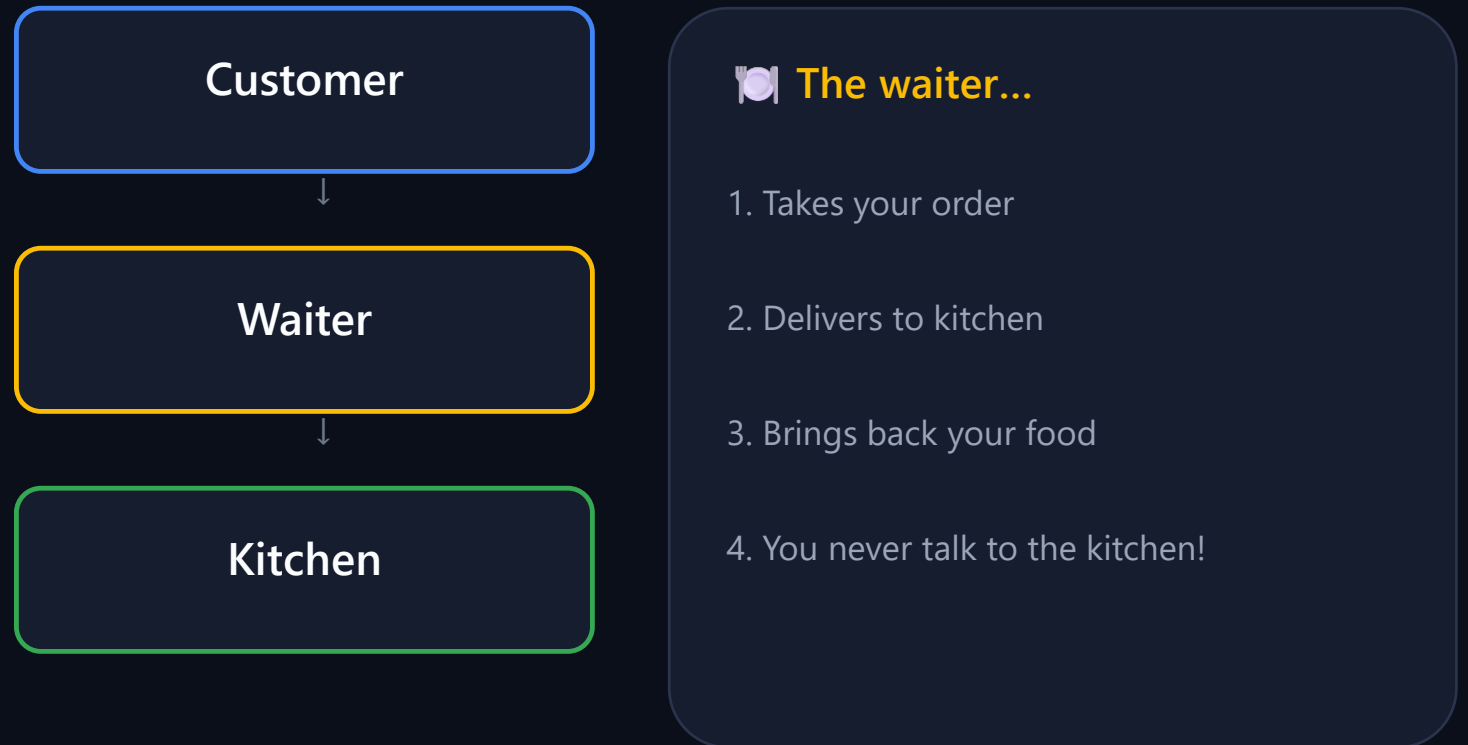
MySQL, PostgreSQL

NoSQL — flexible documents

MongoDB, DynamoDB

Rule of thumb: relational data with relationships → SQL · flexible JSON blobs → NoSQL

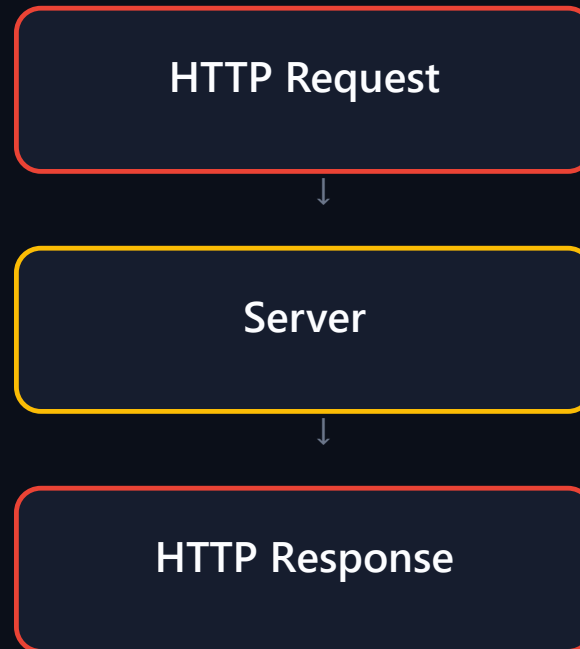
APIs — The Restaurant Analogy



API = Messenger — it carries your request to the system and returns the result

HTTP

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Key idea

Stateless communication — every request is independent, carrying all the info it needs.

Port 80 (HTTP)

Port 443 (HTTPS)

HTTP Methods

Method	Purpose	What it does
GET	Retrieve	Read data
POST	Create	Add new resource
PUT	Replace	Full update
PATCH	Update	Partial update
DELETE	Remove	Delete resource

JSON

example.json

```
{  
  "name": "John",  
  "age": 20,  
  "city": "Visakhapatnam",  
  "skills": ["AI", "Web"]  
}
```

Why JSON everywhere?

- ✓ Language-independent
- ✓ Human-readable
- ✓ Native to JavaScript
- ✓ APIs send/receive JSON
- ✓ Key-value pairs

The universal data format — almost every API returns JSON

Status Codes

200

OK

201

Created

400

Bad Request

401

Unauthorized

403

Forbidden

404

Not Found

500

Internal Server Error

Authentication

API Keys

auth

Simple token for machine clients

Bearer Tokens

auth

In the Authorization header

JWT

auth

Signed, stateless identity token

OAuth

auth

Delegated access — sign in with Google

Webhooks vs Polling

Polling (❌ wasteful)

Ask repeatedly:
"Any updates? Any updates?"

Client pulls every X seconds

Webhook (✅ efficient)

Server calls you when something happens

GitHub

Stripe

Discord

WhatsApp



SECTION 3

Large Language Models

The engine behind the modern AI stack.



What is an LLM?

Large Language Model

A neural network trained on massive amounts of text. It predicts the next token — and that simple skill powers chat, code, and reasoning.

Popular models

GPT OpenAI

Claude Anthropic

Gemini Google

Llama Meta — open weights

How LLMs Work



💡 Tokens

Text is split into tokens — small chunks of characters (~4 chars each). "Hello world" ≈ 2 tokens.

Pricing & limits are measured in tokens

Context Window

Conversation memory — the tokens the model can "see" at once

8K

Small models

32K

Standard

128K

Large context

1M

frontier-class

A bigger context window = the model can read your whole codebase, doc, or conversation

LLM Providers

OpenAI

GPT-4o family

Anthropic

Claude family

Google

Gemini family

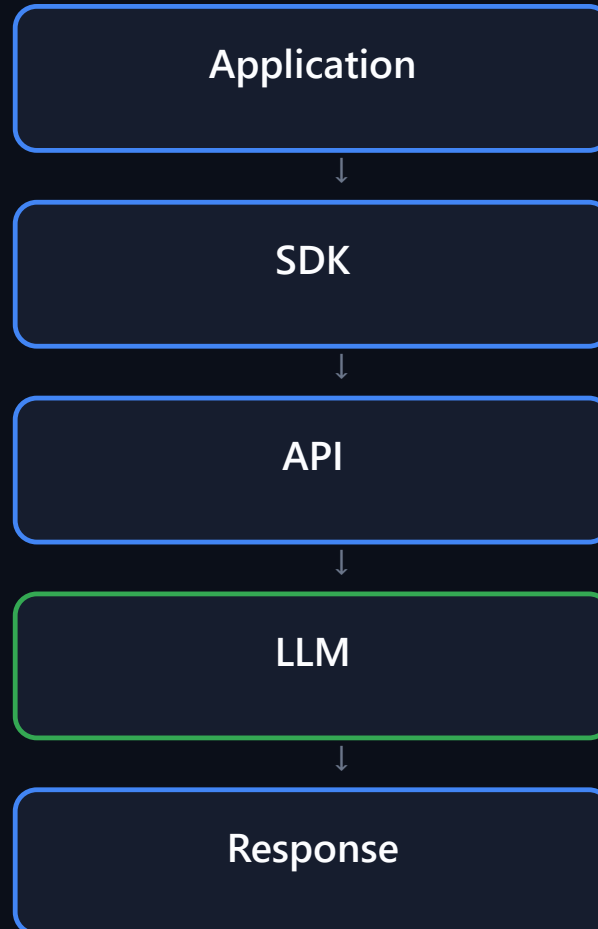
Groq

Ultra-fast inference

OpenRouter

One API, many models

API Workflow



⚙️ What is an SDK?

A library that wraps the API in your language — Python, JS, etc.

```
pip install openai
```

```
npm i @anthropic-ai/sdk
```

OpenAI-Compatible APIs

The magic trick ✨

Write code once with the OpenAI SDK. Switch providers by changing 3 things:

- **Base URL**
- **API Key**
- **Model**

Providers that work

Groq

OpenRouter

Together

Fireworks

DeepSeek

Mistral

Same interface, different prices & speeds

Anthropic APIs

Messages API

chat completions with roles

System Prompt

top-level instructions — the "constitution"

Streaming

token-by-token responses via SSE

Tool Use

functions Claude can call

Function Calling

The model doesn't just talk — it decides to call your tools and uses the results.

Weather

Get forecast

Calendar

Book meetings

Database

Query users

Payment

Charge customer

This is the foundation of AI agents — models + tools = action

Structured Outputs

JSON Schema

```
{
  "type": "object",
  "properties": {
    "title": {"type": "string"},
    "price": {"type": "number"},
    "tags": {"type": "array"}
  },
  "required": ["title"]
}
```

Why it matters

- ✓ Model always returns valid JSON
- ✓ No parsing surprises
- ✓ Type-safe for your app
- ✓ Great for building agents & pipelines

Streaming

Normal response

- ✗ You wait...
- ✗ Server generates everything
- ✗ Full response at once

Streaming

- ✓ First token in ~200ms
- ✓ Token-by-token delivery
- ✓ Feels like the AI is typing



SECTION 4

AI-Assisted Development

Your new workflow: think, prompt, review, ship.



Modern Development Workflow



⚡ AI turbocharges each step

1. Idea → validate fast
2. Research → summarize
3. Design → generate UI
4. Develop → pair-program
5. Deploy → automate

ChatGPT



Use cases

- Research & explain anything
- Planning & project scoping
- Write PRDs & specs
- Learn new topics fast

OpenAI's assistant

Claude



Use cases

- Architecture design
- Long documents & codebases
- Refactoring legacy code
- Deep reasoning

Anthropic's assistant

Gemini

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Use cases

- ▶ Google ecosystem integration
- ▶ Vision & image understanding
- ▶ Multimodal inputs
- ▶ 100+ languages

Google's assistant

Anti Gravity

Agentic Coding

AI writes multi-file changes

Repository Understanding

Indexes your whole codebase

Planning

Breaks tasks into steps

Editing

Inline suggestions & edits

Pro tip: let the agent plan first, then review each change before accepting

OpenCode

CLI

AI coding in your terminal

Providers

Bring your own model

MCP

Connect any MCP server

Skills

Reusable workflows

Terminal

Run commands for you

Google AI Studio

API Keys

Free tier, generous limits

Playground

Chat & test prompts instantly

Prompt Testing

Compare outputs & tune

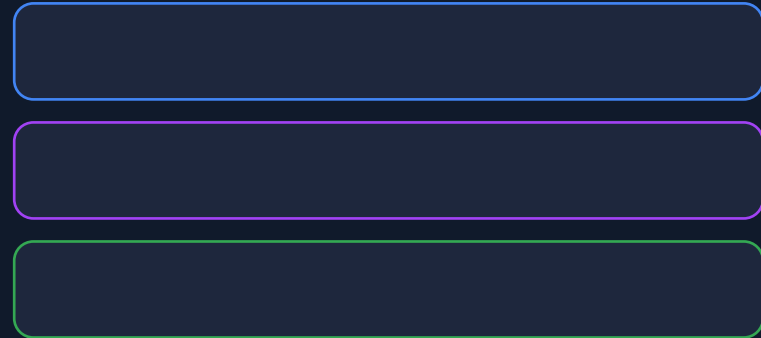
aistudio.google.com — free API keys, no credit card

Stitch

Type a prompt...

"Build a landing page for my AI startup —
dark theme, hero section, pricing cards"

...get a UI ✨

Three stacked rectangular boxes representing a UI prototype. The top box has a blue border, the middle box has a purple border, and the bottom box has a green border. They are all empty, representing different sections of a landing page.

Rapid prototyping — describe it, generate it, iterate

Supabase

Authentication

Email, OAuth, magic links

Database

PostgreSQL with Row-Level Security

Storage

Files & images with policies

Realtime

Live subscriptions for chat/feeds

The open-source Firebase — free tier included

Deployment

Vercel

Frontend & serverless

git push → live 🚀

Cloudflare

Edge, workers, pages

git push → live 🚀

GitHub

Pages, Actions, hosting

git push → live 🚀



SECTION 5

Prompt Engineering

Talk to AI like a senior engineer.



Prompt Engineering

Bad prompt

"make a website"

Vague → generic output

Good prompt

"Build a dark-themed landing page for an AI startup. Include hero, 3 pricing tiers, and a footer. Use React + Tailwind."

Specific → shippable output

Context Engineering

Context

Relevant background & code

Constraints

Budget, time, tech stack

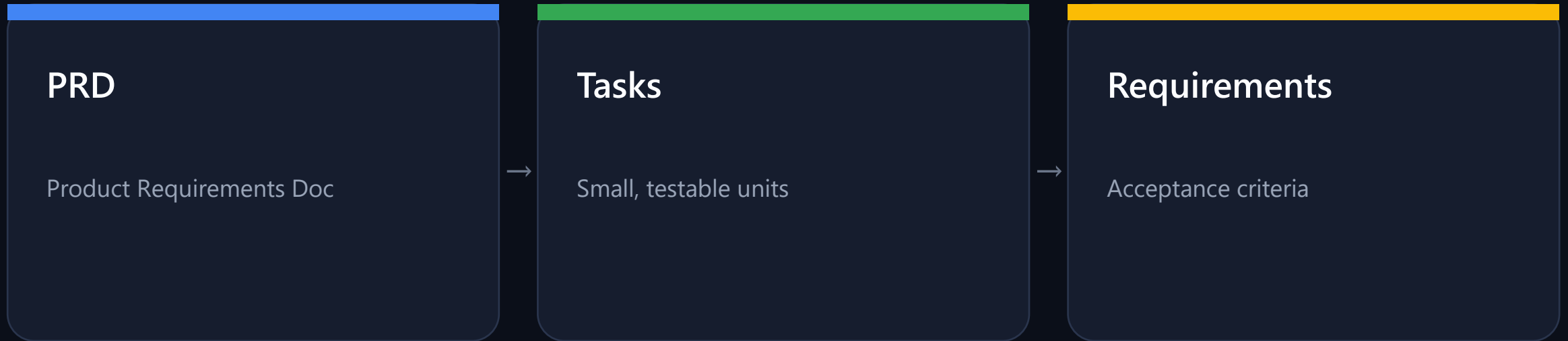
Examples

Show the pattern you want

Output Format

JSON, markdown, file list

Specification-Driven Development



AI reads the spec and builds — you review the diff

AI Code Review

Review

Bugs, edge cases, style

Debug

Explain errors in plain words

Refactor

Clean up with confidence

"Review this diff before I push" — a habit that catches real bugs

Debugging with AI

What to paste

1. The full error message
2. Relevant code snippet
3. The logs / stack trace
4. What you expected vs got

The best prompt

"Here's my code and the error. Explain the root cause, then fix it step by step. Also tell me how to prevent it next time."



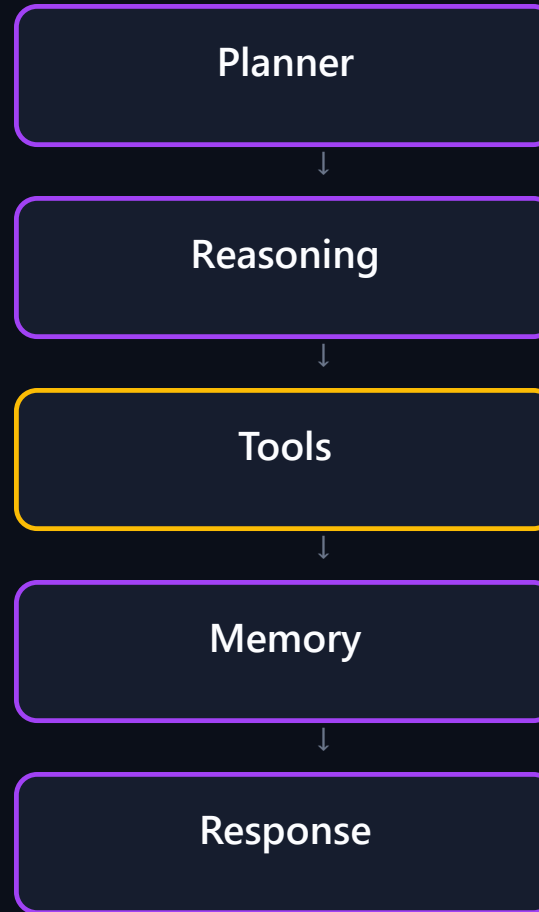
SECTION 6

AI Agents

From chatbots to autonomous workers.



What is an AI Agent?



 **Agent =**

LLM + tools + memory + a loop that keeps going until the task is done.

Not one answer —
a workflow that executes

Agent Architecture

Planner

Breaks the goal into steps

Executor

Runs each step with tools

Memory

Remembers context & results

Reflection

Evaluates & self-corrects

MCP — Model Context Protocol

The problem

LLMs only understand text. They can't touch files, databases, or tools.

The fix

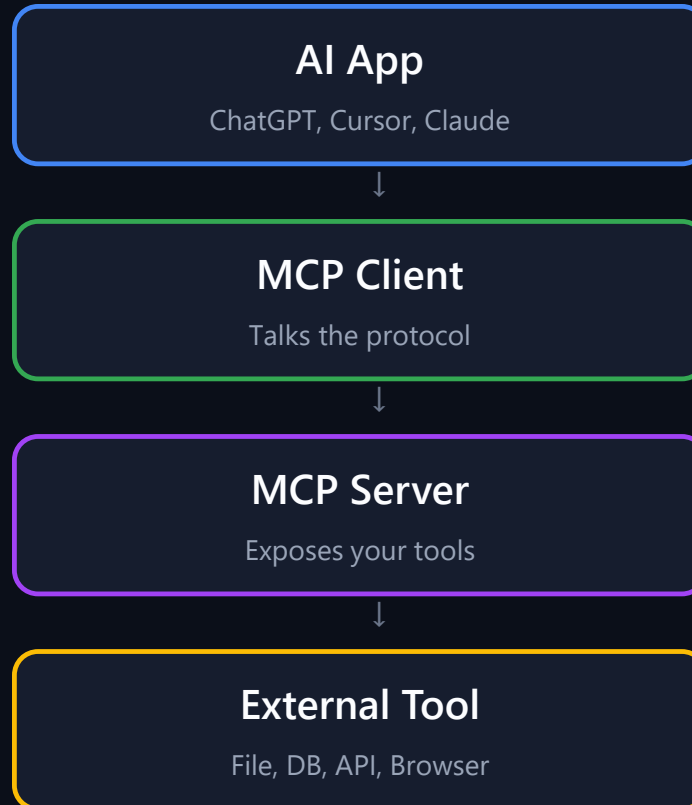
MCP standardizes how AI connects to tools — one protocol, any tool.

MCP components

- Client — The AI app (Cursor, Claude, ChatGPT)
- Server — Exposes tools to the AI
- Resources — Data the AI can read
- Tools — Actions the AI can take
- Prompts — Reusable prompt templates

MCP in Action

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The flow

1. You ask the AI something
2. AI decides it needs a tool
3. MCP Client calls the server
4. Server executes the tool
5. Result comes back to the AI

MCP Ecosystem

Clients (AI apps)

- Cursor — AI-first code editor
- Claude Desktop — Anthropic's assistant
- ChatGPT — Developer Mode connectors
- OpenCode — Terminal-native AI
- Windsurf — Agentic IDE

Registries (find servers)

- Glama — MCP server directory
- Smithery — One-click installs
- mcp.so — Community registry
- PulseMCP — News + directory
- GitHub — The source of truth

Write a server once — every MCP client can use it

Popular MCP Servers

PowerPoint

Create & edit decks

Filesystem

Read & write files

GitHub

Repos, issues, PRs

Browser

Browse & extract

Database

Query PostgreSQL

Terminal

Run commands

Every server here follows the same protocol — connect any of them to your AI

Skills

Packaged workflows your AI agent can invoke — like plugins for agents.

Deploy Website

Ship in one command

Generate Docs

Auto-document code

Git Workflow

Commit, branch, PR

Code Review

Pre-push checks

Tool Calling

Filesystem

read/write files

Browser

browse & extract

GitHub

repos & PRs

Database

query data

Terminal

run commands

More tools = more things your agent can do for you



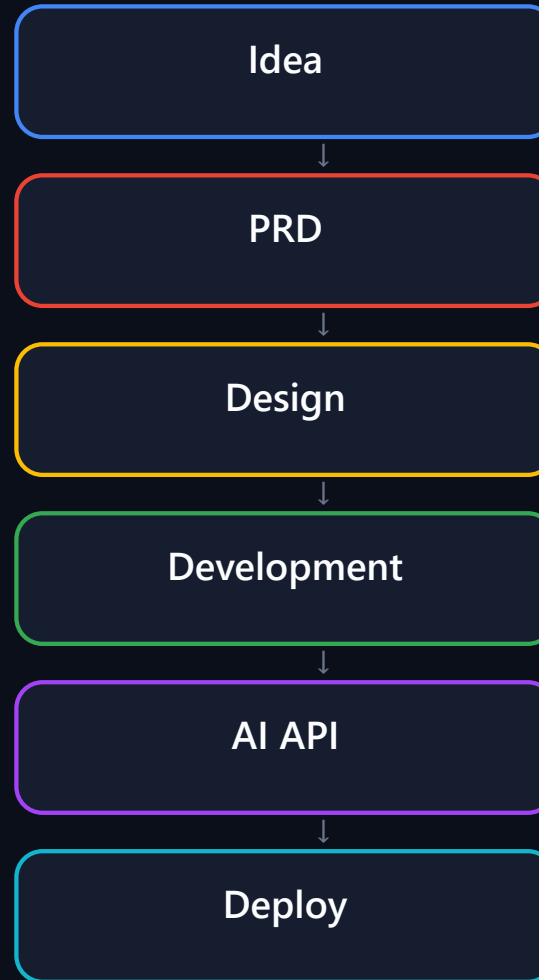
SECTION 7

Building

Everything comes together — demo time.



Complete Workflow



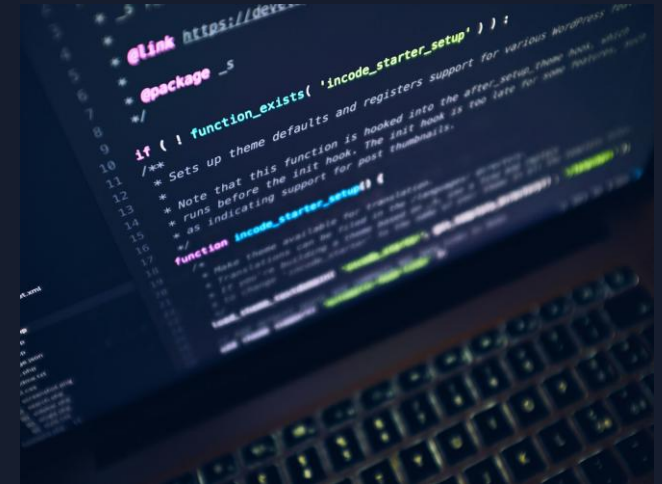
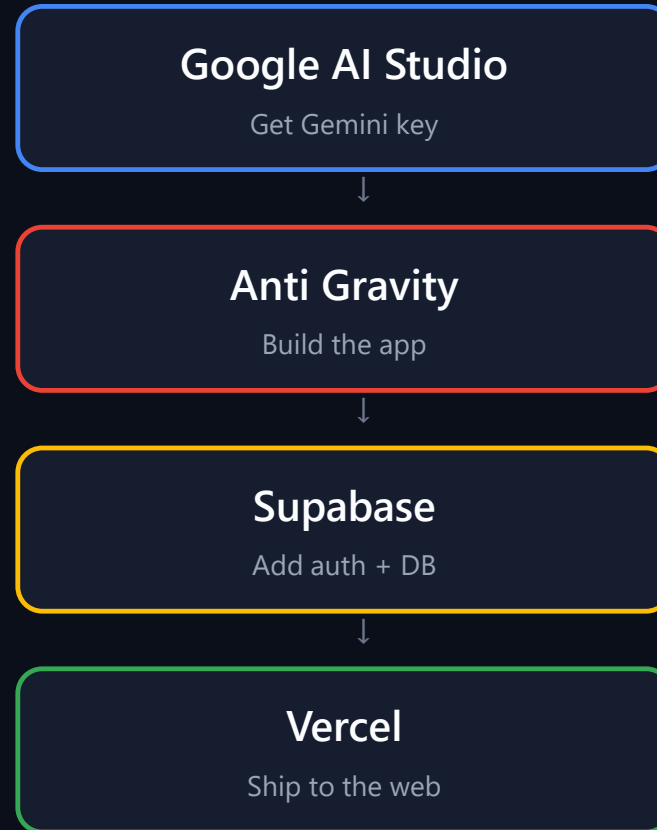
Toolchain

1. ChatGPT / Claude — plan
2. Anti Gravity — build
3. MCP — connect tools
4. Supabase — backend
5. Vercel — deploy

WATCH IT HAPPEN

Live Demo

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A real app, built live with AI in minutes

Performance

Chrome DevTools

Inspect & profile in-browser

Lighthouse

Audit score: perf, SEO, a11y

Network

Watch every request

Caching

Serve faster, spend less

Aim: Lighthouse 90+ · first paint < 1s · cache everything static



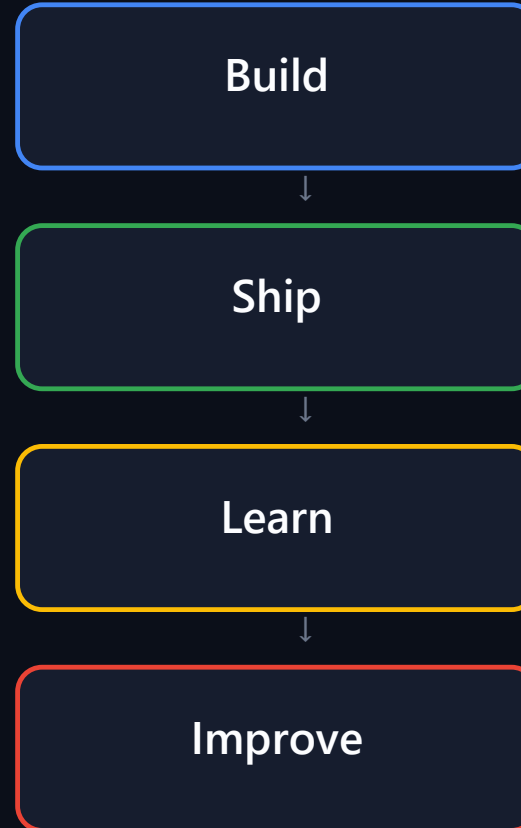
SECTION 8

Startup Mindset

You have the tools. Now go build.



Startup Mindset



Why it works

1. Each loop = real feedback
2. Faster loops = faster learning
3. AI makes loops ~10x faster
4. Small wins compound daily

Product Thinking

Write Code

Features first

What can I build?

Solve Problems

Users first

What problem does this solve?

Startups win by solving real problems — code is just the tool

Career Roadmap



AI-native edge

1. Ship 10 projects
2. Automate your workflow
3. Build with agents
4. Document everything

Resources

MDN

Web platform docs

OpenAI Docs

GPT & API guides

Anthropic Docs

Claude documentation

Google AI Studio

Gemini playground

Supabase Docs

Backend quickstart

Vercel Docs

Deploy in minutes

Thank You

Now go build something.



Team Stero Sonic Labs